A Validated Intrinsic Kernel for a Negative Parity Resonance

Weighted-Riesz Factor Correction, Complement Schur Closure, and Adaptive Spectral Deflation

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Abstract

The negative resonance of a folded-Gaussian Markov operator at the first quadratic band-merging parameter has been isolated in both L^∞ and L^2, with a dimension-uniform Euclidean contour for full and sparse midpoint matrices. The remaining task in the gauge-free peripheral program is to turn that isolated eigenvalue into a validated intrinsic kernel that can be subtracted before trace and determinant analysis. We carry this out at fixed noise $\sigma = 10^{-2}$.

For the parity circle

$$c = -0.9865481927458079, \qquad r = 0.05,$$

we define

$$\mathcal{Q}_-(T;\Gamma) = \frac{1}{2\pi i}\int_\Gamma z(z-T)^{-1}\,dz.$$

An orthogonal Schur decomposition against the full cell-average complement improves the continuum L^2 resolvent upper from 266.650 to 112.95503061584434. The weighted term $\mathcal{Q}_-(\mathcal{K};\Gamma)$ is rank one and has a real smooth Hilbert–Schmidt kernel $q_-(x,y)$. We give explicit Hilbert–Schmidt bounds for q_- and its derivatives through $\partial_x^2\partial_y^2 q_-$, and enclose it in an $L^2([0,1]^2)$ ball of radius 2.427176139794529 about the lifted archived 4096 factor.

A new two-sided residual theorem converts an approximate left–right factor into an exact weighted-Riesz certificate by a block-diagonalizing perturbation. Applied with componentwise outward arithmetic, it proves that the stored parity factors at dimensions 2048, 4096, and 8192 are genuine spectral terms, with Euclidean errors at most 3.66×10^{-10}, 7.21×10^{-10}, and 1.43×10^{-9}. Consequently the two archived Haar transitions are now spectral, not merely algebraic: their actual ratios lie within 3.94×10^{-4} of the quarter–half targets. A smooth-kernel Haar theorem explains these limits intrinsically.

The improved continuum bound yields an all-dimension theorem from $n = 65536$: full and fixed/adaptive sparse exact-real Markov matrices have one parity eigenvalue in the circle and uniform Euclidean resolvent upper 267.81252084743886. Their full-to-sparse weighted-Riesz difference is at most $7.275887086007192 \times 10^{-10}$, while the corresponding intrinsically deflated operators differ by at most $7.277844418892396 \times 10^{-10}$. The full weighted term converges to the midpoint sample of q_- at order n^{-2}, and adaptive support preserves the rate $O(n^{-2}(\log n)^{-1/4})$. No Perron rank-two completion, zero-noise limit, arithmetic trace formula, zeta-zero identification, self-adjoint Hilbert–Pólya operator, or Riemann-hypothesis claim is made.

1 Introduction

Spectral subtraction is straightforward for a normal matrix with an orthonormal eigenbasis: one removes the relevant eigenvalue times its orthogonal projector. The noisy quadratic transfer operators studied here are nonnormal. Their left and right modes carry arbitrary scale, sign, and ordering conventions, and those conventions become increasingly fragile under grid refinement. The invariant object is instead the weighted Riesz term

$$\mathcal{Q}(T;\Gamma) = T\mathcal{P}(T;\Gamma) = \frac{1}{2\pi i}\int_{\Gamma} z(z-T)^{-1}\,dz,$$

where $\mathcal{P}$ is the Riesz projection. For a simple real eigenvalue λ this equals $\lambda r\ell^T/(\ell^T r)$, but the contour formula does not choose a gauge.

The analytic weighted-Riesz framework of Wang [12] proved that a simple isolated smooth Nyström branch has a second-order intrinsic continuum bridge. It also showed that an adaptive Gaussian support cutoff passes through the same calculus under a dimension-uniform Euclidean contour bound. At that stage three statements were still conditional: the negative continuum resonance had not been proved simple and isolated, the required Euclidean conditioning was absent, and the stored factors used in the Haar ledger were known to be spectral only through floating residuals.

The first two conditions were subsequently closed. A validated continuum Grushin–Rouché argument isolated one real negative resonance [6]; a Hilbert–Galerkin lift then established the same circle in L^2 and controlled every sufficiently fine full and sparse midpoint family in the Euclidean norm [8]. The present paper uses those results rather than opening another contour. Its purpose is to construct the intrinsic parity kernel itself and to close the exact-stored factor gate.

Four additional ideas are needed.

1. A two-sided residual correction makes a chosen rank-one factor exact for a nearby matrix, after which contour stability compares it with the true weighted Riesz term.
2. The dyadic Schur formulas can be integrated around the contour. This bounds the change of the weighted term block by block, much more sharply than a direct perturbation by the whole fine operator.
3. The final finite-rank-to-continuum passage should also be Schur, with the full orthogonal complement as the detail space. Its self-energy is quadratic in the mesh and is far smaller than the direct Galerkin defect.
4. The quarter–half law has a continuum explanation. Four midpoint samples in a coarse cell give leading tensors $q_{xx}+q_{yy}$, q_x, q_y, and q_{xy} in the four Haar blocks.

These steps produce an intrinsic deflation

$$\mathcal{K}_\perp = \mathcal{K} - \mathcal{Q}_-(\mathcal{K};\Gamma)$$

that removes the negative branch exactly while leaving the remaining spectrum unchanged away from zero. This is the operator that later trace and determinant papers can use without carrying a left–right gauge.

Three evidence levels remain separate throughout.

Analytic theorem
: Two-sided factor correction, weighted contour stability, infinite complement Schur continuation, smooth-kernel Haar limits, Nyström convergence, and intrinsic deflation.

Outward certificate
: Exact stored binary64 bordered residuals, Arb Frobenius identities, the Hilbert derivative envelope, and every explicit Neumann or Schur gate.

Displayed center
: The heat map in figure 1 is the exact archived 4096 factor center. It is not a pointwise interval enclosure of the continuum kernel.

2 Operator, contour, and main results

Let u_c be the unique root in $(1.5, 1.6)$ of

$$u^3 - 2u^2 + 2u - 2 = 0,$$

fix $\sigma = 1/100$, and put

$$m(x) = 1 - u_c x^2.$$

For $0 \le x, y \le 1$, define

$$g(x,y) = e^{-(y-m(x))^2/(2\sigma^2)} + e^{-(y+m(x))^2/(2\sigma^2)},$$

$$Z(x) = \int_0^1 g(x,y)\, dy, \qquad k(x,y) = \frac{g(x,y)}{Z(x)},$$

and the compact Markov operator

$$(\mathcal{K}f)(x) = \int_0^1 k(x,y) f(y)\, dy \quad \text{on } L^2([0,1]). \tag{1}$$

The parity contour is

$$c = -0.9865481927458079, \qquad r = 0.05, \qquad \Gamma = \{z \in \mathbb{C} : |z - c| = r\}. \tag{2}$$

Write

$$\rho_\Gamma = 0.9365481927458077, \qquad R_\Gamma = 1.036548192745808.$$

For any bounded operator T with $\Gamma \subset \rho(T)$, define

$$\mathcal{P}_-(T;\Gamma) = \frac{1}{2\pi i} \int_\Gamma (z - T)^{-1}\, dz, \tag{3}$$

$$\mathcal{Q}_-(T;\Gamma) = \frac{1}{2\pi i} \int_\Gamma z(z - T)^{-1}\, dz = T\mathcal{P}_-(T;\Gamma). \tag{4}$$

For $n \ge 2$, let $h = 1/n$ and $x_i = y_i = (i + \frac{1}{2})h$. The continuum- normalized midpoint matrix and exactly discretely normalized Markov matrix are

$$(M_n)_{ij} = hk(x_i, y_j), \qquad (P_n)_{ij} = \frac{g(x_i, y_j)}{\sum_q g(x_i, y_q)}. \tag{5}$$

Let $P_n^{(8)}$ be the archived fixed eight-sigma truncation and let

$$L_n = \max\{8, 2\sqrt{\log n}\} \tag{6}$$

define the adaptive sparse family $P_n^{(L_n)}$ as in Wang [7, 8].

Theorem 2.1 (Validated intrinsic continuum parity kernel)**.** *The circle* (2) *lies in the L^2 resolvent set of $\mathcal{K}$ and*

$$\sup_{z\in\Gamma} \left\| (z - \mathcal{K})^{-1} \right\|_{2\to 2} \le 112.95503061584434. \tag{7}$$

The operator

$$Q_- = \mathcal{Q}_-(\mathcal{K};\Gamma)$$

has rank one and a real C^∞ kernel $q_-(x,y)$. Its Hilbert–Schmidt norm lies in

$$0.9365481927458077 \le \|q_-\|_{L^2([0,1]^2)} \le 5.854166642320046.$$

The derivative uppers are recorded in table 3.

Let $\widehat{Q}_{4096}$ be the stored binary64 factor

$$\widehat{Q}_{4096} = c\,\frac{r_0 \ell_0^T}{\ell_0^T r_0}$$

lifted to the cell space V_{4096}. Then

$$\left\|Q_- - \widehat{Q}_{4096}\right\|_{2\to 2} \le 1.716272707582898, \tag{8}$$

$$\|q_- - \widehat{q}_{4096}\|_{L^2([0,1]^2)} \le 2.427176139794529. \tag{9}$$

Thus the continuum weighted term is a validated intrinsic kernel, not only an eigenvalue count.

Theorem 2.2 (Three exact stored parity factors)**.** *For each $n \in \{2048, 4096, 8192\}$, let A_n^{s} be the exact stored binary64 Markov matrix, let c_n, r_n, ℓ_n be its archived parity data, and put*

$$\widehat{Q}_n = c_n \frac{r_n \ell_n^T}{\ell_n^T r_n}.$$

The circle $|z - c_n| = 0.05$ contains exactly one eigenvalue of A_n^{s}. Its true weighted Riesz term Q_n^{s} obeys

$$\left\|Q_n^{\mathrm{s}} - \widehat{Q}_n\right\|_2 \le \begin{cases} 3.650836314181483 \times 10^{-10}, & n = 2048, \\ 7.202868434316492 \times 10^{-10}, & n = 4096, \\ 1.426439999332510 \times 10^{-9}, & n = 8192. \end{cases}$$

Consequently the actual spectral Haar ratios satisfy the intervals in table 2. Every interval is within 10^{-3} of its smooth- kernel quarter–half target.

Theorem 2.3 (Uniform full, sparse, and deflated families)**.** *For every integer $n \ge 65536$, each of*

$$M_n, \qquad P_n, \qquad P_n^{(8)}, \qquad P_n^{(L_n)}$$

has exactly one eigenvalue inside Γ, counted algebraically. It is real, negative, and simple. Uniformly in n,

$$\sup_{z\in\Gamma} \left\|(z - M_n)^{-1}\right\|_2 \le 267.3688881197896, \tag{10}$$

$$\sup_{z\in\Gamma} \left\|(z - P_n)^{-1}\right\|_2 \le 267.8125208334001, \tag{11}$$

$$\sup_{z\in\Gamma} \left\|(z - P_n^{(8)})^{-1}\right\|_2, quad \sup_{z\in\Gamma} \left\|(z - P_n^{(L_n)})^{-1}\right\|_2 \le 267.81252084743886. \tag{12}$$

Moreover,

$$\left\|\mathcal{Q}_-(P_n^{(8)};\Gamma) - \mathcal{Q}_-(P_n;\Gamma)\right\|_2, \quad \left\|\mathcal{Q}_-(P_n^{(L_n)};\Gamma) - \mathcal{Q}_-(P_n;\Gamma)\right\|_2 \le 7.275887086007192 \times 10^{-10}. \tag{13}$$

Define the intrinsic deflations

$$P_{n,\perp} = P_n - \mathcal{Q}_-(P_n;\Gamma), \qquad P_{n,\perp}^{(L)} = P_n^{(L)} - \mathcal{Q}_-(P_n^{(L)};\Gamma).$$

Then

$$\left\|P_{n,\perp}^{(8)} - P_{n,\perp}\right\|_2, \quad \left\|P_{n,\perp}^{(L_n)} - P_{n,\perp}\right\|_2 \le 7.277844418892396 \times 10^{-10}. \tag{14}$$

Corollary 2.4 (Unconditional parity weighted-kernel bridge)**.** *Let*

$$(Q_n^\circ)_{ij} = hq_-(x_i, y_j).$$

Then

$$\|\mathcal{Q}_-(P_n;\Gamma) - Q_n^\circ\|_2 = O(n^{-2}), \tag{15}$$

$$\left\|\mathcal{Q}_-(P_n^{(L_n)};\Gamma) - Q_n^\circ\right\|_2 = O\Big(n^{-2}(\log n)^{-1/4}\Big). \tag{16}$$

Thus every hypothesis of the conditional parity theorem in Wang [12] is now proved. No eigenvector gauge appears in either statement.

3 Two-sided residual correction of a weighted factor

The first new ingredient closes the gap between a stored low-rank factor and the actual spectral term of the stored matrix. It is purely algebraic and applies to any nonnormal matrix.

Proposition 3.1 (Rank-one block-diagonalizing correction)**.** *Let $A \in \mathbb{C}^{n\times n}$ and let $\lambda_0 \in \mathbb{C}$. Choose nonzero vectors r_0, ℓ_0 with*

$$g = \ell_0^T r_0 \neq 0, \qquad P_0 = \frac{r_0\ell_0^T}{g}.$$

Put $R_0 = A - \lambda_0 \mathrm{I}$ and

$$\Delta = -R_0P_0 - P_0R_0 + P_0R_0P_0. \tag{17}$$

Then

$$\widetilde{A} = A + \Delta = \lambda_0 P_0 + (\mathrm{I} - P_0)A(\mathrm{I} - P_0), \tag{18}$$

and

$$\widetilde{A}P_0 = P_0\widetilde{A} = \lambda_0 P_0.$$

If

$$u = R_0 r_0, \qquad v^T = \ell_0^T R_0, \qquad p = \frac{\|r_0\|_2\,\|\ell_0\|_2}{|g|},$$

then

$$\|\Delta\|_2 \leq \frac{\|u\|_2\,\|\ell_0\|_2}{|g|} + \frac{\|r_0\|_2\,\|v\|_2}{|g|} + p\min\left\{\frac{\|u\|_2\,\|\ell_0\|_2}{|g|}, \frac{\|r_0\|_2\,\|v\|_2}{|g|}\right\}. \tag{19}$$

Proof. Since $P_0^2 = P_0$, expansion of $\lambda_0P_0 + (\mathrm{I} - P_0)A(\mathrm{I} - P_0) - A$ gives (17). The invariance identities follow immediately. Moreover,

$$R_0P_0 = \frac{u\ell_0^T}{g}, \qquad P_0R_0 = \frac{r_0v^T}{g}.$$

Their spectral norms are the first two terms of (19). Finally,

$$\|P_0R_0P_0\|_2 \leq \min\{\|P_0\|_2\,\|R_0P_0\|_2\,, \|P_0R_0\|_2\,\|P_0\|_2\},$$

which gives the third term. □

Theorem 3.2 (Validated weighted factor)**.** *Let Γ be a circle of radius r and maximum modulus R_Γ. Suppose A has algebraic count one inside Γ and*

$$\sup_{z\in\Gamma}\left\|(z-A)^{-1}\right\|_2 \leq M.$$

Assume λ_0 lies inside Γ and let ε be the right side of (19). If $M\varepsilon < 1$, then

$$\|\mathcal{Q}_-(A;\Gamma) - \lambda_0P_0\|_2 \leq rR_\Gamma\frac{M^2\varepsilon}{1-M\varepsilon}. \tag{20}$$

In particular, λ_0P_0 is a validated center for the actual weighted Riesz term.

Proof. By proposition 3.1, $\left\|\widetilde{A}-A\right\|_2 \le \varepsilon$. The Neumann lemma keeps Γ in the resolvent set of $\widetilde{A}$, with resolvent upper $M/(1-M\varepsilon)$, and preserves the algebraic count. The projection P_0 commutes with $\widetilde{A}$, its range carries the eigenvalue λ_0, and the total count is one. Hence

$$\mathcal{Q}_-(\widetilde{A};\Gamma) = \lambda_0 P_0.$$

The first resolvent identity in the contour formula (4) now gives (20); this is the weighted Lipschitz theorem of Wang [12] with all constants displayed. ☐

Remark 3.3 (A simultaneous eigenvalue enclosure). The Grushin certificate gives, throughout its disk,

$$\left|E_{-+}(z) + \frac{z-c_n}{g_n}\right| \le \varepsilon_{*,n}.$$

The maximum principle extends the boundary bound to the interior. At the unique zero,

$$|\lambda_n - c_n| \le |g_n|\varepsilon_{*,n}.$$

The resulting uppers at 2048, 4096, and 8192 are respectively 1.782×10^{-13}, 3.475×10^{-13}, and 6.849×10^{-13}.

4 Exact stored factors and the spectral quarter–half law

The stored matrices come from the RH-36 and RH-37 snapshots [9, 10]. For each parity center, a sparse LU approximation to the bordered inverse is recomputed. Every residual row and column is then evaluated against the exact stored graph using componentwise outward arithmetic [5, 11]. The inequality

$$\|X\|_2 \le \sqrt{\|X\|_1 \, \|X\|_\infty}$$

converts the exact 1- and ∞-norm ledgers into Euclidean bounds.

Table 1: Exact stored Euclidean Grushin and factor-correction ledger.

n	$\|R\|_2$	contour resolvent	$\|\Delta\|_2$	$M\,\|\Delta\|_2$	$\left\|Q_n^{\mathrm{s}} - \widehat{Q}_n\right\|_2$
2048	$4.262{\times}10^{-11}$	83.92380	$1.001{\times}10^{-12}$	$8.394{\times}10^{-11}$	$3.651{\times}10^{-10}$
4096	$1.181{\times}10^{-10}$	84.00733	$1.970{\times}10^{-12}$	$1.655{\times}10^{-10}$	$7.203{\times}10^{-10}$
8192	$3.296{\times}10^{-10}$	84.03470	$3.898{\times}10^{-12}$	$3.276{\times}10^{-10}$	$1.427{\times}10^{-9}$

All three correction products are below 4×10^{-10}. Thus the archived parity factors are not merely small-residual diagnostics: they are centers of exact spectral weighted terms with explicit Euclidean radii.

Let $\mathscr{H}_{2n}$ be the orthogonal pairwise Haar transform. Write

$$\mathscr{H}_{2n} Q_{2n}^{\mathrm{s}} \mathscr{H}_{2n}^T = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}, \qquad E_n = A_n - Q_n^{\mathrm{s}}.$$

The Arb factor algebra from RH-40 gives exact Frobenius intervals for the same blocks built from $\widehat{Q}_n$ [4, 12]. If $e_n = \left\|Q_n^{\mathrm{s}} - \widehat{Q}_n\right\|_2$, then every compressed error has rank at most two, and therefore

$$\left\|B_n - \widehat{B}_n\right\|_{\mathrm{F}}, \left\|C_n - \widehat{C}_n\right\|_{\mathrm{F}}, \left\|D_n - \widehat{D}_n\right\|_{\mathrm{F}} \le \sqrt{2}\, e_{2n},$$

while

$$\left\|E_n - \widehat{E}_n\right\|_{\mathrm{F}} \le \sqrt{2}(e_n + e_{2n}).$$

This converts the exact-factor ledger into actual spectral intervals.

Table 2: Validated ratios of actual spectral parity Haar blocks.

block	rigorous ratio interval	smooth target	maximum deviation
E	$[0.2497729080,\ 0.2498601648]$	$1/4$	2.271×10^{-4}
C	$[0.5000272443,\ 0.5000292074]$	$1/2$	2.921×10^{-5}
B	$[0.5000320106,\ 0.5000338986]$	$1/2$	3.390×10^{-5}
D	$[0.2496654557,\ 0.2503931293]$	$1/4$	3.932×10^{-4}

This closes the exact-stored spectral-factor limitation stated explicitly in RH-40. It does not validate every future binary64 rebuild; it validates the three archived matrices and factors whose hashes are recorded in the certificate.

5 The smooth intrinsic continuum kernel

Let

$$\Pi_- = \mathcal{P}_-(\mathcal{K};\Gamma), \qquad Q_- = \mathcal{Q}_-(\mathcal{K};\Gamma).$$

The continuum count is one, so Π_- and Q_- have rank one. If λ_- is the enclosed eigenvalue, then

$$Q_- = \lambda_- \Pi_-, \qquad \mathcal{K}\Pi_- = \Pi_-\mathcal{K} = Q_-. \tag{21}$$

Proposition 5.1 (Smooth Hilbert–Schmidt kernel)**.** *The weighted term Q_- has a real C^∞ kernel $q_-(x,y)$. If*

$$M_\Gamma = sup_{z\in\Gamma} \left\| (z-\mathcal{K})^{-1} \right\|_{2\to 2},$$

then

$$\|\Pi_-\|_{2\to 2} \le rM_\Gamma, \qquad \|Q_-\|_{2\to 2} \le rR_\Gamma M_\Gamma. \tag{22}$$

Because Q_- has rank one,

$$\|q_-\|_{L^2([0,1]^2)} = \|Q_-\|_{2\to 2}.$$

For source derivatives,

$$\|\partial_x^a q_-\|_{L^2} \le \|\partial_x^a k\|_{L^2} \|\Pi_-\|_{2\to 2},$$

and analogously for pure target derivatives. Mixed derivatives satisfy

$$\left\|\partial_x^a \partial_y^b q_-\right\|_{L^2} \le \rho_\Gamma^{-1} \|\partial_x^a k\|_{L^2} \|\Pi_-\|_{2\to 2} \left\|\partial_y^b k\right\|_{L^2}. \tag{23}$$

Proof. The kernel k is smooth. Since a Hilbert–Schmidt operator composed with a bounded operator remains Hilbert–Schmidt, $Q_- = \mathcal{K}\Pi_-$ has a kernel and source derivatives are obtained by differentiating the left kernel factor. The identity $Q_- = \Pi_-\mathcal{K}$ gives the target derivatives. Repeated differentiation is permitted because every derivative kernel is Hilbert–Schmidt.

The contour length is $2\pi r$, giving (22). Rank one makes the Hilbert–Schmidt and operator norms equal. Finally,

$$\mathcal{K}\Pi_-\mathcal{K} = \lambda_- Q_-,$$

so

$$Q_- = \lambda_-^{-1}\mathcal{K}\Pi_-\mathcal{K}.$$

Differentiate the two outer kernels and use $|\lambda_-| \ge \rho_\Gamma$ to obtain (23). Reality follows from the real operator, the conjugation-invariant circle, and algebraic count one. $\square$

The RH-42 Arb envelope gives the explicit constants in table 3.

Table 3: Validated Hilbert–Schmidt envelope for the intrinsic parity kernel.

quantity	rigorous upper
$\|q_-\|_{L^2}$	5.854166642320046
$\|(q_-)_x\|_{L^2}$	3780.968408944756
$\|(q_-)_y\|_{L^2}$	2160.665606696173
$\|(q_-)_{xx}\|_{L^2}$	$1.107398829423628 \times 10^6$
$\|(q_-)_{xy}\|_{L^2}$	$1.544489270236306 \times 10^6$
$\|(q_-)_{yy}\|_{L^2}$	$2.679665935766761 \times 10^5$
$\|(q_-)_{xxyy}\|_{L^2}$	$5.610208521245975 \times 10^{10}$

Applying the midpoint-average Peano theorem of Wang [8] to q_- itself gives, at $n = 65536$,

$$\left\| Q_n^\circ - Q_n^{\mathrm{G}} \right\|_2 \le 1.790125512353268 \times 10^{-5}, \tag{24}$$

where Q_n^{G} is the cell-average matrix of q_-. Thus the intrinsic kernel has a direct second-order midpoint discretization even before it is compared with the weighted term of P_n.

6 Infinite-complement Schur closure

RH-42 passed from the last Galerkin operator to the continuum through the direct defect $\left\| \mathcal{K} - P_n^{\mathrm{G}} \mathcal{K} P_n^{\mathrm{G}} \right\|$. That estimate is valid but adds the two one-sided errors. The complement can instead be retained as a Schur block, where its influence on the parity branch is quadratic.

Let $P = P_n^{\mathrm{G}}$ be the orthogonal cell-average projection and $Q = \mathrm{I} - P$. Relative to $L^2 = V_n \oplus V_n^\perp$,

$$\mathcal{K} = \begin{pmatrix} A_n & B_n \\ C_n & D_n \end{pmatrix}, \qquad A_n = P\mathcal{K}P. \tag{25}$$

The cellwise Poincaré–Wirtinger inequality gives

$$\|C_n\|_2 \le \frac{K_x}{\pi n}, \qquad \|B_n\|_2 \le \frac{K_y}{\pi n}, \qquad \|D_n\|_2 \le \frac{K_{xy}}{\pi^2 n^2}. \tag{26}$$

These are the same formulas as the dyadic Haar bounds, but the detail space is now the full infinite-dimensional complement.

Theorem 6.1 (Continuum complement Schur step)**.** *Suppose*

$$\sup_{z \in \Gamma} \left\| (z - A_n)^{-1} \right\|_2 \le M,$$

and write b, c, d for the three uppers in (26). If

$$d < \rho_\Gamma, \qquad M b (\rho_\Gamma - d)^{-1} c < 1,$$

then $\mathcal{K}$ and A_n have the same algebraic count inside Γ. The full resolvent is bounded by the Frobenius norm of

$$\begin{pmatrix} S & SbR_d \\ R_d c S & R_d + R_d c S b R_d \end{pmatrix},$$

where

$$R_d = (\rho_\Gamma - d)^{-1}, \qquad S = \frac{M}{1 - M b R_d c}.$$

At $n = 65536$ the Schur product is

$$0.0007281434256176882$$

and the resulting continuum resolvent upper is (7).

Proof. The detail spectrum lies in $|z| \le d$, disjoint from the disk bounded by Γ. Take the Schur complement of $z - D_n$. The self-energy $B_n(z - D_n)^{-1}C_n$ has norm at most bR_dc, so the displayed product gives the Schur inverse and the four block bounds. Homotoping B_n, C_n to zero keeps the detail block fixed and the majorants decrease. At the decoupled endpoint $A_n \oplus D_n$, the detail block contributes no eigenvalue inside Γ, proving count preservation. The argument uses only operator norms and therefore does not require the detail space to be finite-dimensional. $\square$

This improves the previous continuum upper by a factor 2.36. More importantly, it supplies the constants for a sharp weighted-term transport.

7 Weighted Schur transport and the construction ball

The block resolvent formula contains more information than the norm of the full resolvent. Let

$$T = \begin{pmatrix} A & B \\ C & D \end{pmatrix}$$

and suppose the detail block has no spectrum in Γ. Put

$$R_A = (z - A)^{-1}, \quad R_D = (z - D)^{-1}, \quad S = (z - A - BR_DC)^{-1}.$$

Then

$$(z - T)^{-1} = \begin{pmatrix} S & SBR_D \\ R_DCS & R_D + R_DCSBR_D \end{pmatrix}.$$

Since $\mathcal{Q}_-(D;\Gamma) = 0$, subtracting the decoupled weighted term leaves only

$$\begin{aligned} &S - R_A = SBR_DCR_A, \\ &SBR_D, \qquad R_DCS, \qquad R_DCSBR_D. \end{aligned}$$

Proposition 7.1 (Weighted Schur block bound)**.** *Assume uniform bounds*

$$\|R_A\| \le M, \quad \|R_D\| \le R_d, \quad \|S\| \le S_*, \quad \|B\| \le b, \quad \|C\| \le c$$

on Γ. Then

$$\|\mathcal{Q}_-(T;\Gamma) - \operatorname{diag}(\mathcal{Q}_-(A;\Gamma), 0)\|_2$$

is at most the Frobenius norm of

$$rR_\Gamma \begin{pmatrix} S_*bR_dcM & S_*bR_d \\ R_dcS_* & R_dcS_*bR_d \end{pmatrix}. \tag{27}$$

Proof. Insert the four resolvent differences above into (4). The contour length contributes r and the weight contributes R_Γ. A block operator whose block norms are bounded entrywise by a nonnegative 2×2 matrix has norm at most the matrix spectral norm, hence at most its Frobenius norm. $\square$

The construction of theorem 2.1 starts with the validated stored factor at 4096 and applies the weighted Lipschitz or weighted Schur bound at every subsequent gate. The ledger is shown in table 4.

Table 4: Validated operator-norm construction of the continuum weighted term.

gate	weighted-term difference upper
stored factor $\to$ exact stored Q	7.203×10^{-10}
exact stored $\to M_{4096}$	0.003863696504533
$M_{4096} \to A_{4096}$	0.319148940875075
$A_{4096} \to A_{8192}$	0.891990587282860
$A_{8192} \to A_{16384}$	0.312894707103005
$A_{16384} \to A_{32768}$	0.114813078602118
$A_{32768} \to A_{65536}$	0.049768188532663
$A_{65536} \to \mathcal{K}$ complement	0.023793507962359
total	1.716272707582898

The first and final operators both have rank one, so their difference has rank at most two. Therefore its Hilbert–Schmidt norm is at most $\sqrt{2}$ times its operator norm, yielding (9).

8 The smooth quarter–half Haar theorem

The exact stored ratios in table 2 have a simple continuum origin. Let $q \in C^4([0,1]^2)$ and form its midpoint matrix

$$(Q_h^\circ)_{ij} = hq(x_i, y_j).$$

Within each coarse cell, the two fine midpoints are $x_i \pm h/4$ and $y_j \pm h/4$. Choose the detail orientation so that the signed coefficient is $+1$ at the positive offset. The Haar blocks of the fine midpoint matrix have entries

$$(A_h)_{ij} = \frac{h}{4} \sum_{\varepsilon,\eta=\pm 1} q(x_i + \varepsilon h/4, y_j + \eta h/4), \tag{28}$$

$$(C_h)_{ij} = \frac{h}{4} \sum_{\varepsilon,\eta=\pm 1} \varepsilon q(x_i + \varepsilon h/4, y_j + \eta h/4), \tag{29}$$

$$(B_h)_{ij} = \frac{h}{4} \sum_{\varepsilon,\eta=\pm 1} \eta q(x_i + \varepsilon h/4, y_j + \eta h/4), \tag{30}$$

$$(D_h)_{ij} = \frac{h}{4} \sum_{\varepsilon,\eta=\pm 1} \varepsilon\eta q(x_i + \varepsilon h/4, y_j + \eta h/4). \tag{31}$$

Put $E_h = A_h - Q_h^\circ$.

Theorem 8.1 (Intrinsic quarter–half limits)**.** *Under the piecewise-constant Hilbert–Schmidt identification,*

$$h^{-2}E_h \longrightarrow \frac{q_{xx} + q_{yy}}{32}, \tag{32}$$

$$h^{-1}C_h \longrightarrow \frac{q_x}{4}, \tag{33}$$

$$h^{-1}B_h \longrightarrow \frac{q_y}{4}, \tag{34}$$

$$h^{-2}D_h \longrightarrow \frac{q_{xy}}{16} \tag{35}$$

in Hilbert–Schmidt norm. Whenever the corresponding limiting tensor is nonzero, refinement $h \mapsto h/2$ gives norm ratios

$$\frac{1}{4}, \qquad \frac{1}{2}, \qquad \frac{1}{2}, \qquad \frac{1}{4}$$

for E, C, B, D, respectively.

Proof. Taylor-expand the four samples in (31). Symmetry cancels all odd terms in A_h, all terms except those odd in x in C_h, all terms except those odd in y in B_h, and all terms except those odd in both variables in D_h. Entrywise,

$$\begin{aligned} E_h &= \frac{h^3}{32}(q_{xx} + q_{yy}) + O(h^5), \\ C_h &= \frac{h^2}{4} q_x + O(h^4), \\ B_h &= \frac{h^2}{4} q_y + O(h^4), \\ D_h &= \frac{h^3}{16} q_{xy} + O(h^5), \end{aligned}$$

uniformly on the coarse midpoint grid. An $n \times n$ midpoint matrix with entries $hf(x_i, y_j)$ has Frobenius norm converging to $\|f\|_{L^2}$. Divide by the displayed powers of h and use midpoint Riemann convergence. The ratio statement follows if the limiting norm is positive. □

Applied to the intrinsic q_- from theorem 2.1, this theorem explains why the validated stored ratios already sit so close to their quarter–half limits. The theorem is gauge-free: it concerns the kernel q_-, not separately normalized left and right modes.

9 Uniform sparse transfer and intrinsic deflation

The improved continuum resolvent changes the all-grid threshold. Combining the continuum Galerkin defect and the second-order midpoint defect gives, at $n = 65536$,

$$\left\| \mathcal{K} - \widetilde{M}_n \right\|_{2 \to 2} \leq 0.005112929739977326.$$

Its product with (7) is

$$0.5775311353158 < 1,$$

which yields (10). The defect decreases for every larger integer n.

The exact normalizer lower is

$$Z_{\min} \geq 0.012533141373155001.$$

At the threshold, the discrete row-normalization defect is

$$\|P_n - M_n\|_2 \leq 6.19557781914536 \times 10^{-6},$$

with Neumann product 0.0016565047527645261. Finally, the relaxed eight-sigma cutoff theorem gives

$$\left\| P_n^{(8)} - P_n \right\|_2 \leq 1.957332885203986 \times 10^{-13}.$$

Every factor in this fixed-width upper is nonincreasing with n. Since $L_n \geq 8$, the same number controls the adaptive family. This proves the resolvent and count statements in theorem 2.3.

The weighted first-resolvent identity gives

$$\|\mathcal{Q}_-(A; \Gamma) - \mathcal{Q}_-(B; \Gamma)\|_2 \leq r R_\Gamma M_A M_B \|A - B\|_2 .$$

Inserting the full and sparse resolvent uppers proves (13).

Proposition 9.1 (Exact intrinsic deflation). *Let T have one algebraically simple eigenvalue λ inside Γ, and put $Q = \mathcal{Q}_-(T;\Gamma)$. Then*

$$T_\perp = T - Q$$

commutes with the Riesz projection, is zero on its range, and agrees with T on its complementary invariant subspace. Hence

$$\operatorname{spec}(T_\perp) \setminus \{0\} = \operatorname{spec}(T) \setminus \{\lambda\} \tag{36}$$

with algebraic multiplicity away from zero.

Proof. For a simple branch, $Q = \lambda \mathcal{P}_-$. Both T and Q commute with $\mathcal{P}_-$. On $\operatorname{Ran} \mathcal{P}_-$ their difference is zero, while on $\ker \mathcal{P}_-$ the weighted term vanishes. The invariant direct-sum decomposition gives (36). □

For full and sparse matrices,

$$\left\| P_{n,\perp}^{(L)} - P_{n,\perp} \right\|_2 \le \left\| P_n^{(L)} - P_n \right\|_2 + \left\| \mathcal{Q}_-(P_n^{(L)};\Gamma) - \mathcal{Q}_-(P_n;\Gamma) \right\|_2 .$$

This proves (14). Under the adaptive schedule, RH-39 gives

$$\left\| P_n^{(L_n)} - P_n \right\|_2 = O\left(n^{-2} (\log n)^{-1/4} \right),$$

so both the weighted term and the deflated sparse family inherit the same rate relative to their full counterparts.

Finally, RH-40 already proves

$$\| \mathcal{Q}_-(P_n;\Gamma) - Q_n^\circ \|_2 = O(n^{-2})$$

for a simple isolated smooth continuum branch. The continuum simplicity and the uniform Euclidean cutoff condition are now both unconditional, so corollary 2.4 follows from Anselone [1], Atkinson [2], Chatelin [3], Wang [12].

Remark 9.2 (Fixed width versus adaptive width). The fixed eight-sigma weighted difference is uniformly below 7.28×10^{-10}, which is ample for spectral deflation. Nevertheless the underlying row-norm cutoff has a mathematically nonzero continuum floor. Adaptive growth is required when convergence to the full continuum kernel, rather than uniform spectral stability, is the claim.

10 Certificate summary and figure

The complete certificate closes the following gates.

1. exact stored Euclidean Grushin counts at 2048, 4096, and 8192;
2. two-sided correction products below 3.28×10^{-10};
3. actual spectral quarter–half intervals at both stored transitions;
4. an infinite-complement Schur product below 7.29×10^{-4};
5. a continuum weighted-term construction ball;
6. an all-grid exact-real theorem from $n = 65536$;

7. full-to-sparse weighted and deflated differences below 7.28×10^{-10}.

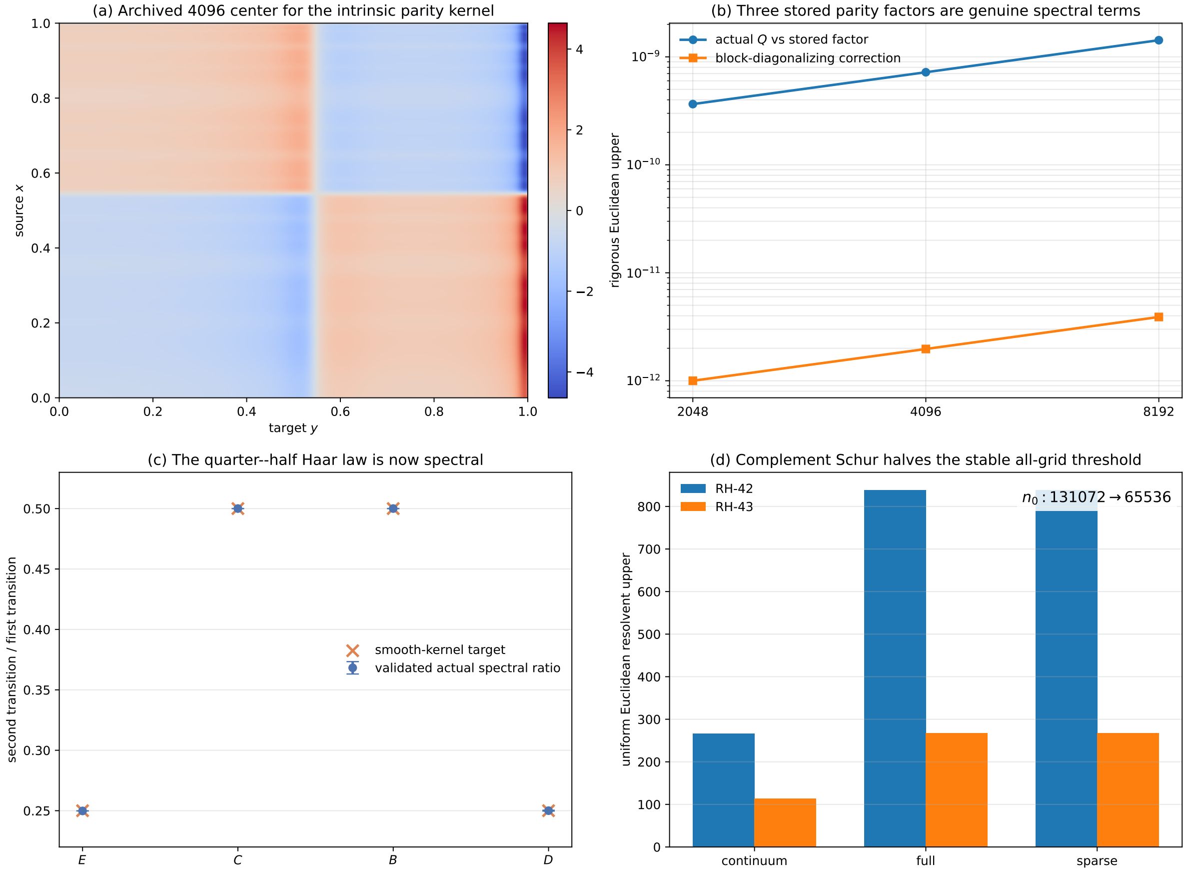

Figure 1: Validated intrinsic parity-kernel closure. (a) The exact archived 4096 factor center, displayed as a piecewise kernel; it is not a pointwise interval enclosure of q_-. (b) The block-diagonalizing corrections and actual weighted-term errors for all three stored factors. (c) Corrected Frobenius intervals for the actual spectral Haar ratios, together with the smooth-kernel quarter–half targets. (d) The continuum complement Schur step improves the continuum and all-grid resolvent constants and halves the stable certified threshold.

11 What is closed and what remains

The present result changes the status of the weighted-Riesz route in four ways.

1. **The negative continuum branch is now an intrinsic kernel.** It is rank one, smooth, gauge-free, and enclosed in operator and Hilbert–Schmidt norms around an archived center.

2. **The stored parity factors are actual spectral terms.** Their previous floating residual status has been replaced by exact stored-matrix Grushin and correction certificates at all three levels.

3. **The quarter–half mechanism is both analytic and spectral.** The continuum theorem identifies the derivative tensors, while the stored ratio intervals now concern the true matrix weighted terms.

4. **Parity deflation is ready for later trace work.** The operator $\mathcal{K}_\perp$ and its matrix analogues remove the negative eigenvalue without a mode normalization convention.

Several boundaries are equally important.

1. The noise width remains fixed at $\sigma = 1/100$.
2. The continuum kernel is enclosed in L^2 and derivative Hilbert–Schmidt norms; no pointwise interval heat map is claimed.
3. The all-dimension theorem concerns exact-real Gaussian formulas. The binary64 theorem concerns only the three archived matrices.
4. The Perron weighted term has not been merged with the parity term. A validated rank-two intrinsic deflation is the natural next gate.
5. No self-adjoint generator, $T \log T$ law, arithmetic trace formula, prime-power identity, zeta-zero identification, or Riemann-hypothesis conclusion follows from this paper.

12 Archive and reproducibility

The archive contains two principal ledgers:

1. the multilevel exact-stored Euclidean Grushin certificate; and
2. the complete factor, Haar, complement, kernel, family, cutoff, and deflation composition certificate.

Every cross-paper input, local source, result, figure, and publication artifact is recorded by SHA-256.

The complete replay is:

```
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/run_multilevel_euclidean_grushin.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_weighted_kernel_certificate.py
MPLBACKEND=Agg PYTHONDONTWRITEBYTECODE=1 \
  /root/math/.venv/bin/python experiments/make_figures.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  -m pytest -q -p no:cacheprovider
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
cp main.pdf validated-weighted-riesz-parity-kernel.pdf
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/build_archive.py
PYTHONDONTWRITEBYTECODE=1 /root/math/.venv/bin/python \
  experiments/verify_archive.py
```

The 8192 bordered inverse replay is the longest step, taking roughly six minutes on the current server. The composition, Arb Haar ledger, tests, and figures take only seconds.

13 Conclusion

An isolated eigenvalue is not yet a usable spectral subtraction. The appropriate object is its weighted Riesz term. Here that term has been constructed as a validated smooth continuum kernel, connected by an explicit chain to the archived 4096 factor, and transferred uniformly to full and adaptive sparse matrix families.

The same argument resolves two older ambiguities. Two-sided residuals now prove that all three archived parity factors are genuine spectral terms, and the quarter–half Haar law follows from both a smooth-kernel theorem and corrected actual-matrix intervals. Treating the infinite orthogonal complement by Schur rather than by a direct norm defect improves the continuum conditioning enough to halve the stable all-grid threshold.

The resulting deflated operator removes the negative branch exactly and preserves the rest of the spectrum away from zero. This is the first strictly intrinsic continuum object in the route that can be handed to a subsequent trace or determinant construction. The next step is not an arithmetic identification; it is the validated addition of the Perron term and the formation of a rank-two intrinsic complement.

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